

Overall methodology (four-module closed loop)

$\mathcal{G} + \mathcal{M}$ (offline artifacts)

User
instruction u
+ context c

A. KG Construction

Schema + normalization
Constraints interface

B. Intent & Params

Self-reflect intent
Slot completion (KG)

C. Template Synthesis

Retrieve + match
Rule-guided rendering

D. Verification

Syntax + safety + semantics
Self-correction loop

G-code y
+ report r

feedback / repair

